

FACTORY SIMULATION — FINAL REPORT

OOP with C++ GIST EECS

1. Team Information

Team name	BBANG
Factory domain	Smart Farm Bakery
GitHub repository	https://github.com/eh-rud/BBANG.git

Member	Student ID	Primary responsibility
김도경	20255021	Leader & Frontend Architecture & Interface Design (Developing Dear ImGui UI layouts, prototyping View layers, and implementing UI-to-Controller Command/Snapshot DTO bindings)
황서현	20255243	Backend Core Logic & State Machine (Designing the Factory Simulation Object hierarchy, implementing concrete machine process timers, and managing core update loops)

2. Class Inventory

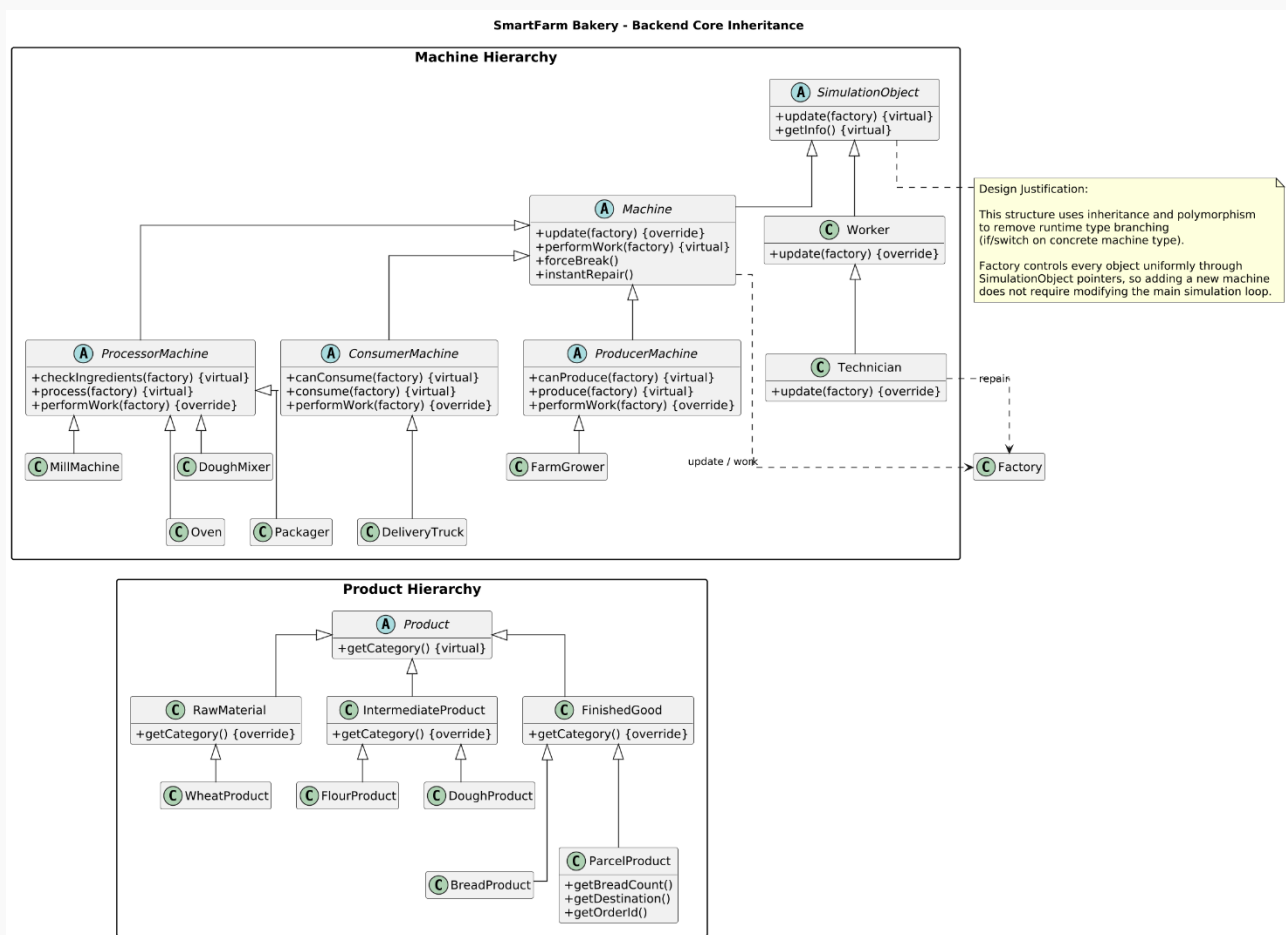
Class name	Layer (Model/View/Controller)	Abstract / Concrete	Inherits from	Status
SimulationObject	Model	Abstract	-	Done
Machine	Model	Abstract	SimulationObject	Done
ProducerMachine	Model	Abstract	Machine	Done
ProcessorMachine	Model	Abstract	Machine	Done
ConsumerMachine	Model	Abstract	Machine	Done
FarmGrower	Model	Concrete	ProducerMachine	Done
MillMachine	Model	Concrete	ProcessorMachine	Done
DoughMixer	Model	Concrete	ProcessorMachine	Done
Oven	Model	Concrete	ProcessorMachine	Done
Packager	Model	Concrete	ProcessorMachine	Done
DeliveryTruck	Model	Concrete	ConsumerMachine	Done
Worker	Model	Abstract	SimulationObject	Done
Technician	Model	Concrete	Worker	Done
Product	Model	Abstract	-	Done

RawMaterial	Model	Concrete base class	Product	Done
IntermediateProduct	Model	Concrete base class	Product	Done
FinishedGood	Model	Concrete base class	Product	Done
WheatProduct	Model	Concrete	RawMaterial	Done
FlourProduct	Model	Concrete	IntermediateProduct	Done
DoughProduct	Model	Concrete	IntermediateProduct	Done
BreadProduct	Model	Concrete	FinishedGood	Done
ParcelProduct	Model	Concrete	FinishedGood	Done
FarmPlot	Model	Concrete	-	Done
StorageBin	Model	Concrete	-	Done
OrderItem	Model	Concrete	-	Done
EventItem	Model	Concrete	-	Done
Factory	Model / Controller	Concrete	-	Done
QueueInfo	Model / View DTO	Struct	-	Done
FactoryCommand	Controller / View DTO	Struct	-	Done
FactorySnapshot	Model / View DTO	Struct	-	Done
PlotSnapshot	Model / View DTO	Struct	-	Done
StorageSnapshot	Model / View DTO	Struct	-	Done
OrderSnapshot	Model / View DTO	Struct	-	Done
EventSnapshot	Model / View DTO	Struct	-	Done
MachineSnapshot	Model / View DTO	Struct	-	Done
TruckLoadSnapshot	Model / View DTO	Struct	-	Done
UIState	View	Struct	-	Done
Theme	View	Namespace / UI helper module	-	Done
UI	View	Namespace / reusable UI component module	-	Done
frontend.cpp module	View	Module	-	Done
UI_Core.cpp module	View	Module	-	Done
UI_Scenes.cpp module	View	Module	-	Done
main.cpp module	Controller / Application	Module	-	Done

ZoneTab	View / Model	Enumeration	-	Done
PlotState	Model	Enumeration	-	Done
MachineState	Model	Enumeration	-	Done
Scenario	Model / Controller	Enumeration	-	Done
ItemType	Model	Enumeration	-	Done
EventType	Model	Enumeration	-	Done

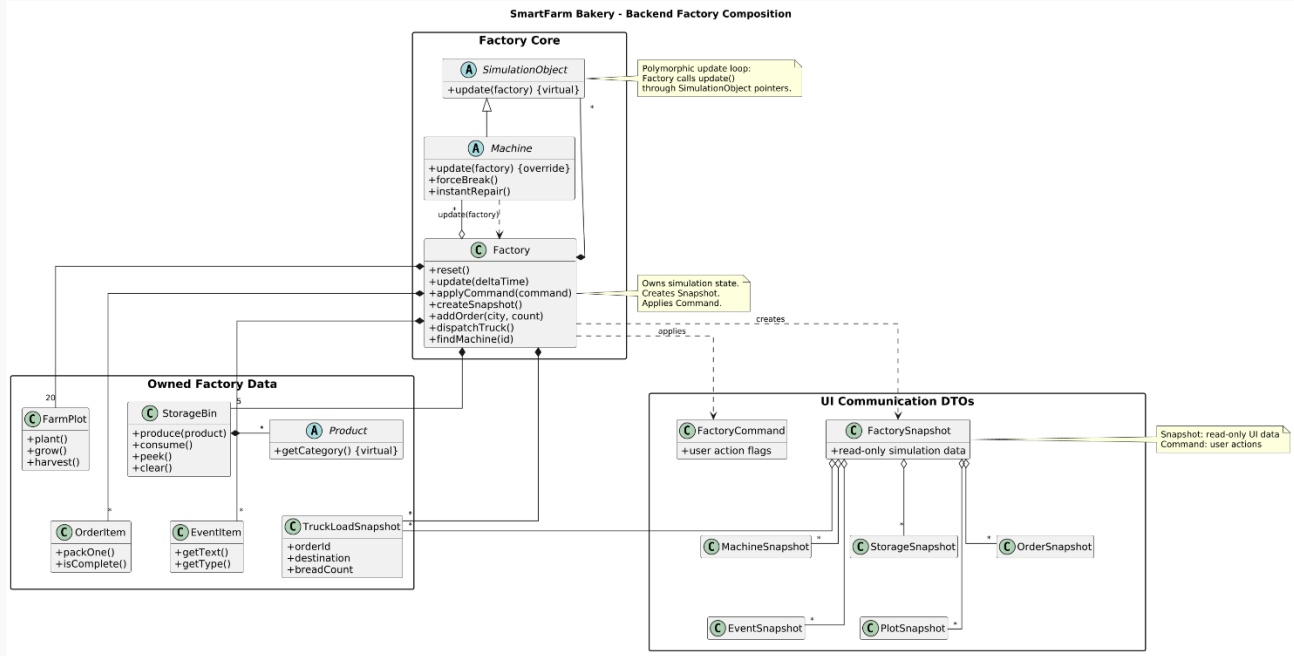
3. Final Class Diagrams and Architecture Diagrams

1. Backend Core Inheritance UML



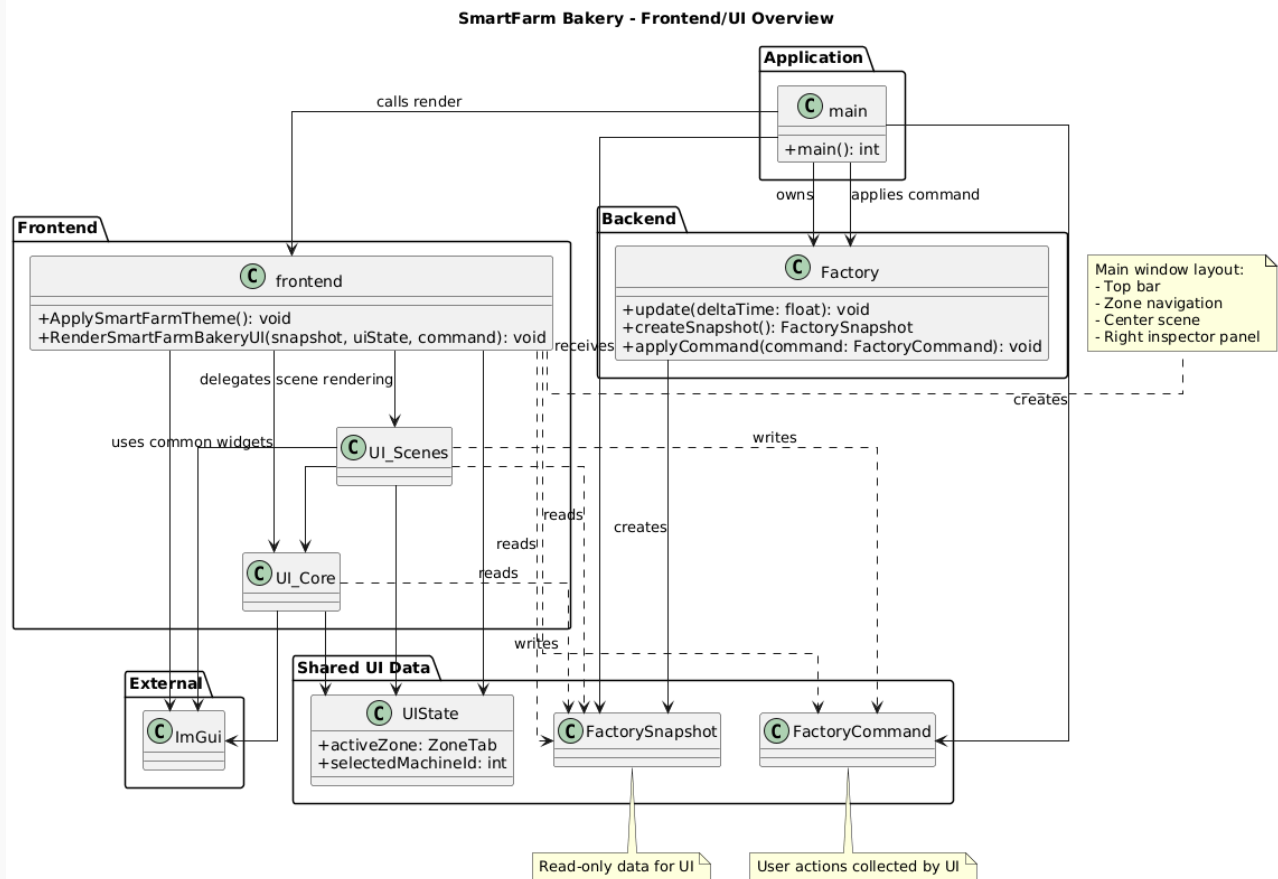
A class hierarchy of machines and products that achieves perfect polymorphism by utilizing SimulationObject as the base class, completely eliminating runtime type branching (if/switch statements).

2. Backend Factory Composition UML



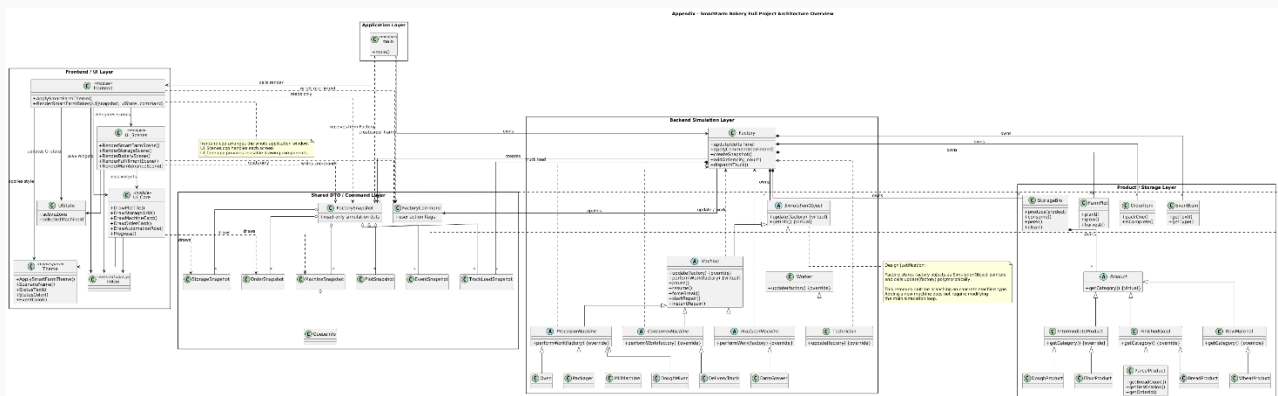
An architectural view illustrating how the Factory core centrally manages the simulation state and owned data, while generating read-only Snapshot and write-only Command objects to communicate with the UI.

3. Frontend/UI Overview UML



A modular rendering architecture completely isolated from the business logic, utilizing ImGui and reusable components (UI_Core) to visualize simulation data and capture user commands.

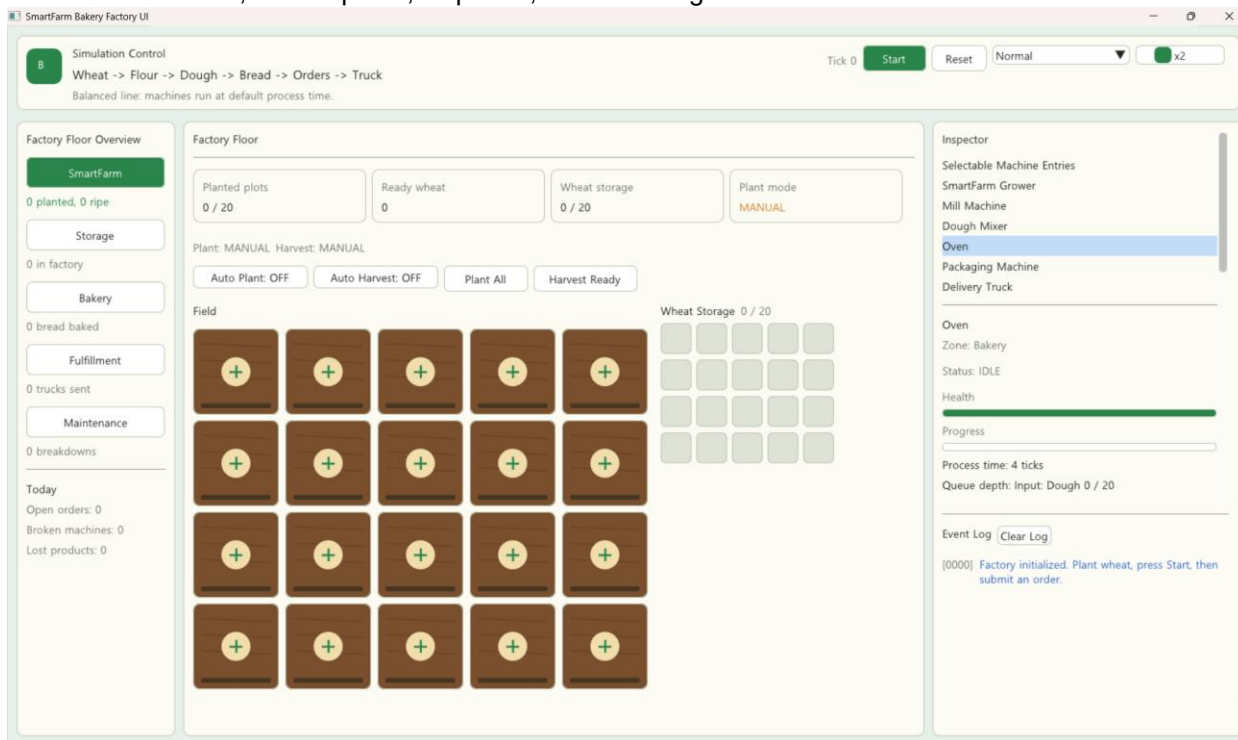
4. Full Project Overview UML



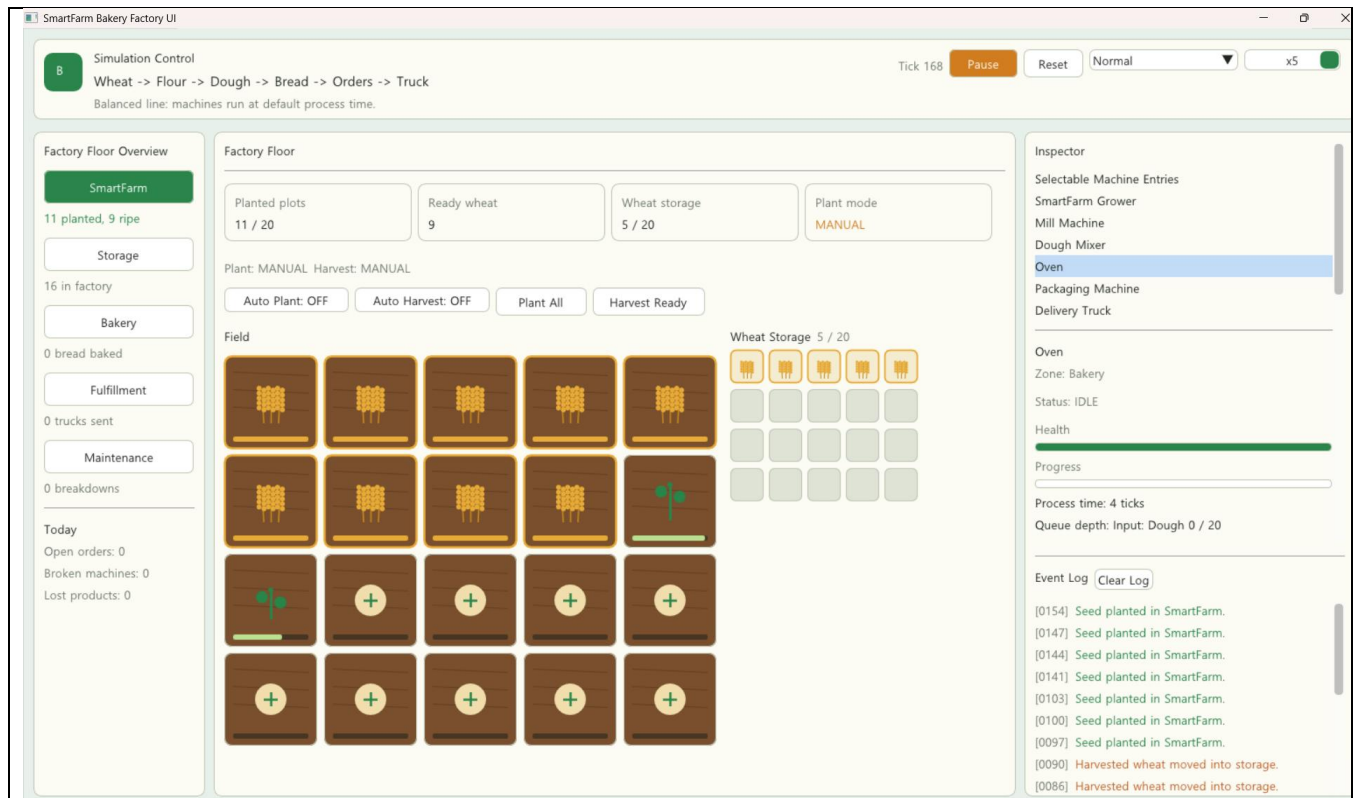
A comprehensive blueprint of the system demonstrating loose coupling between the application (main), frontend, backend, and data layers via Data Transfer Objects (DTOs) without direct dependencies.

4. UI Screenshots

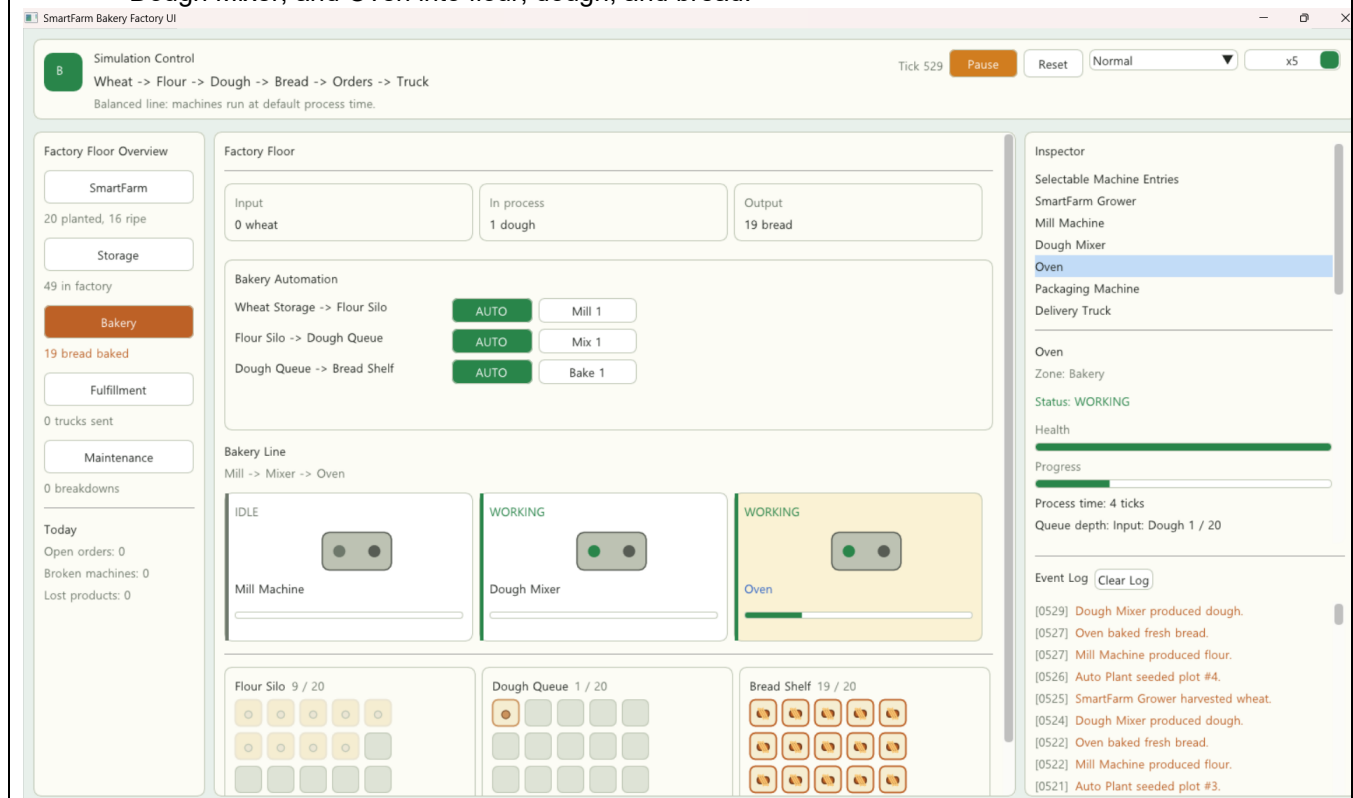
1. Initial screen of the SmartFarm Bakery simulation before any action is taken, showing the default dashboard, control panel, inspector, and event log.



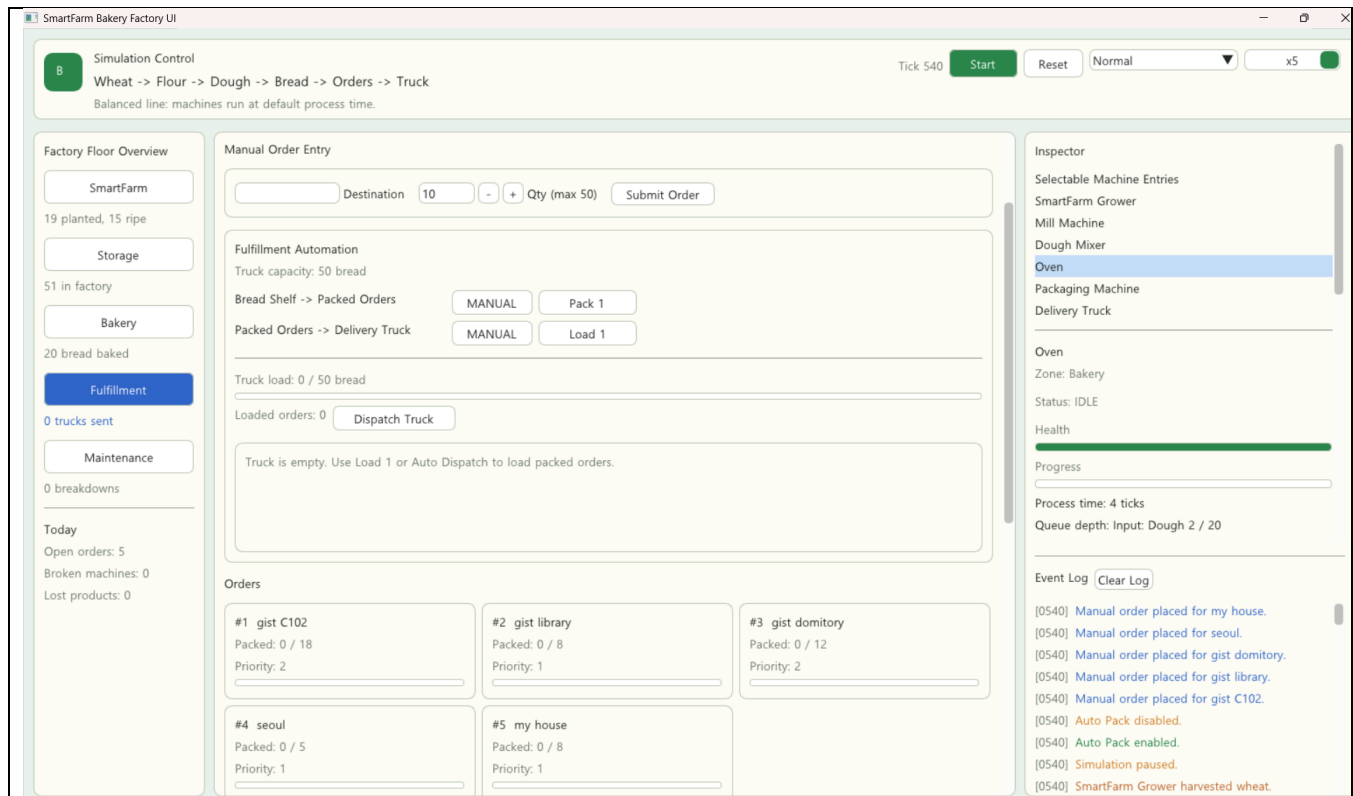
2. SmartFarm view after planting seeds and harvesting wheat, showing plot growth status and harvested wheat stored in Wheat Storage.



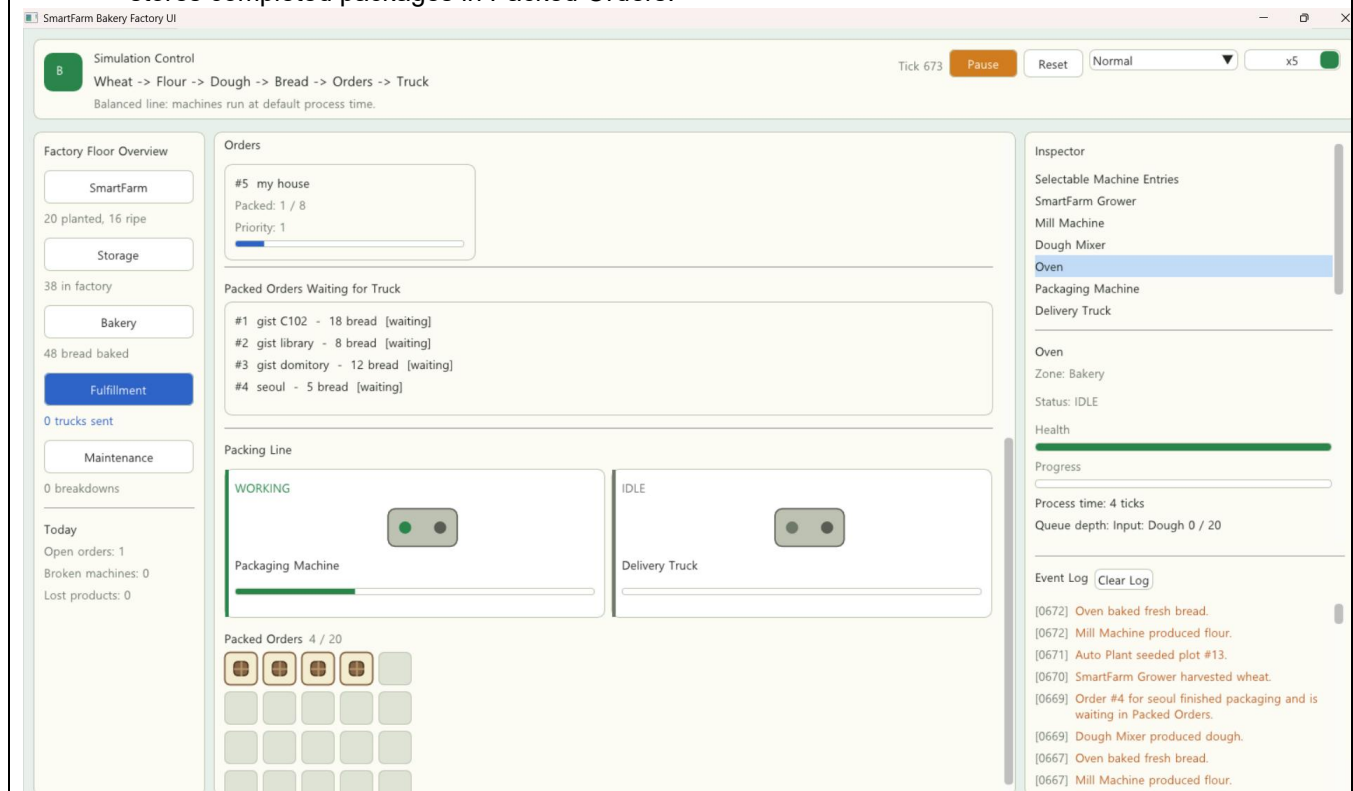
3. Bakery view showing the production flow in which harvested wheat is processed by the Mill Machine, Dough Mixer, and Oven into flour, dough, and bread.



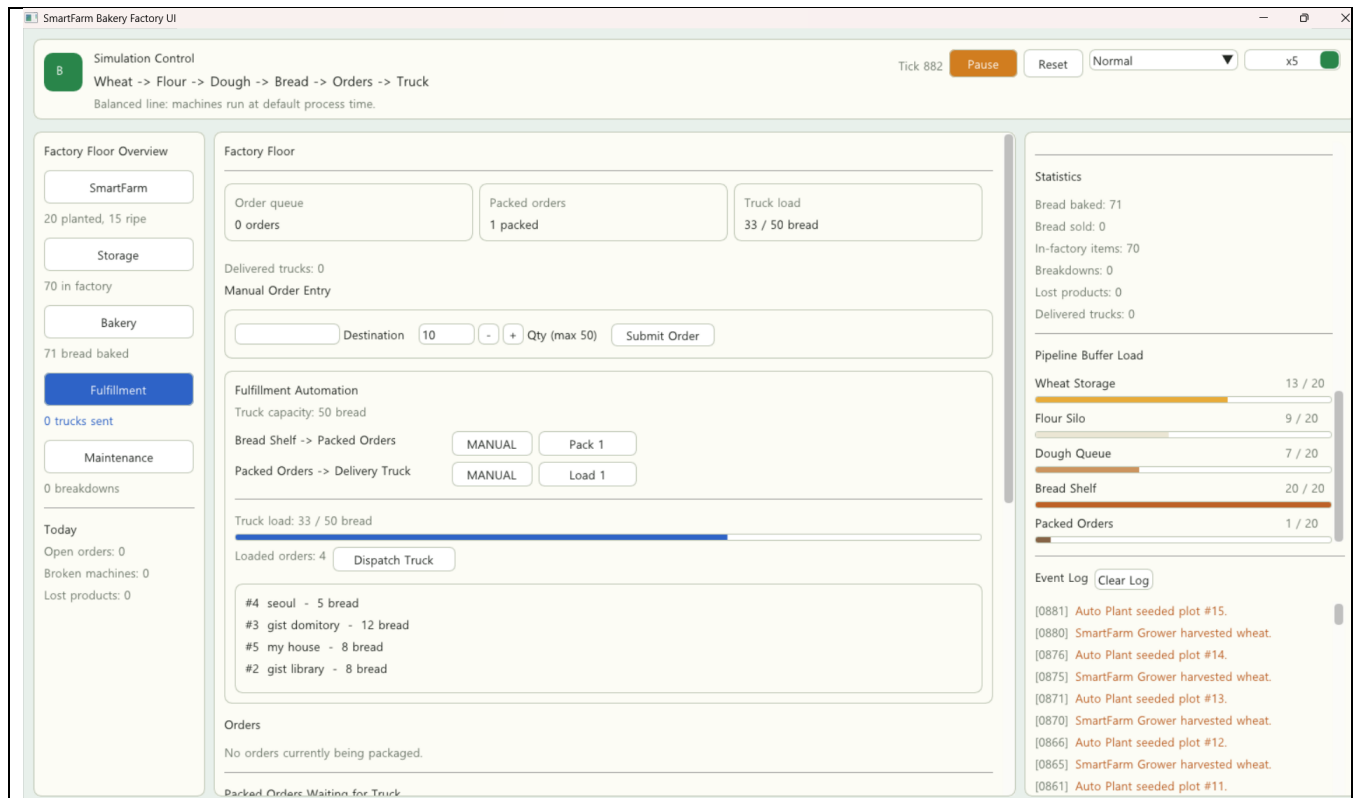
4. Fulfillment view after multiple customer orders have been entered, displaying the active order list and fulfillment control panel.



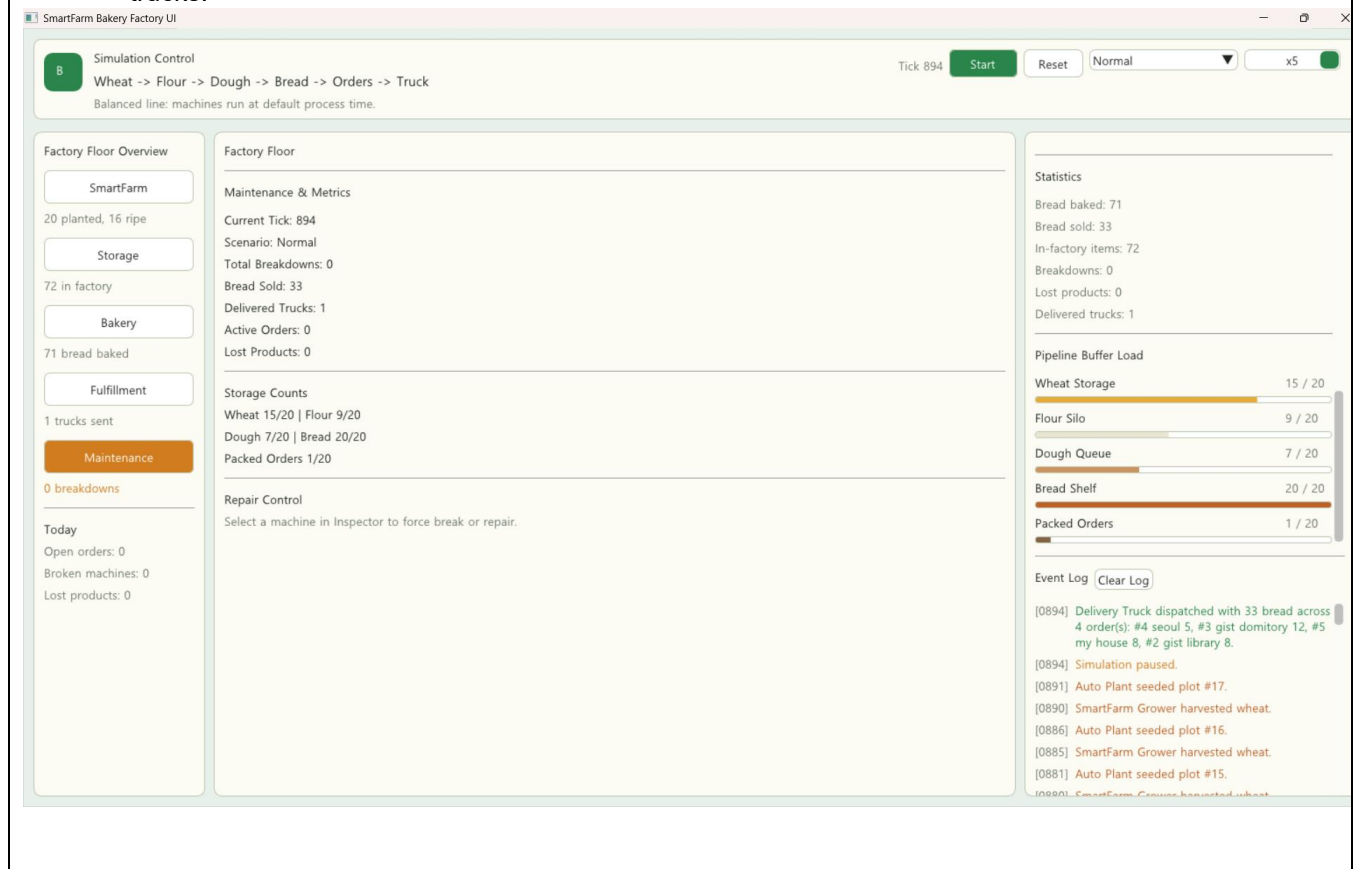
5. Fulfillment process in which the Packaging Machine packs bread according to incoming orders and stores completed packages in Packed Orders.



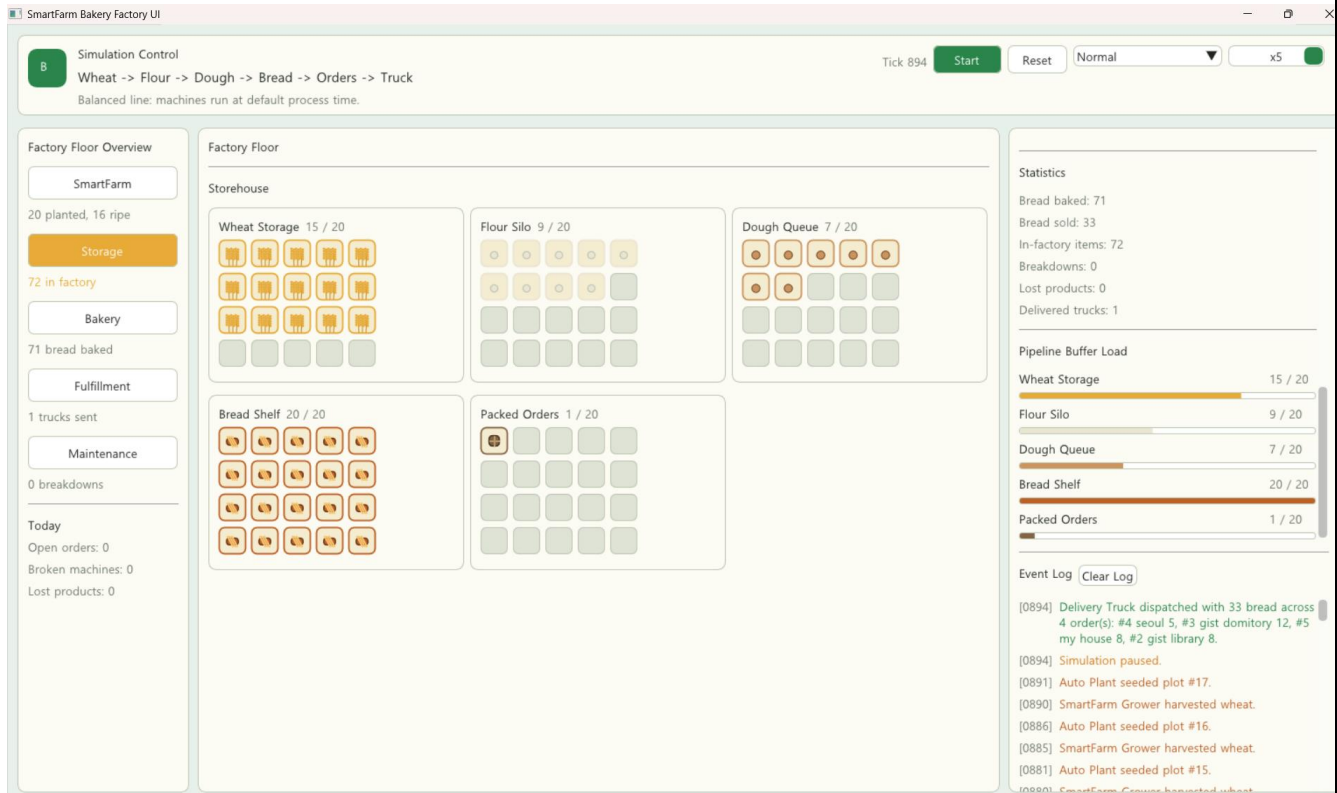
6. Fulfillment view showing packed orders loaded onto the Delivery Truck before dispatch.



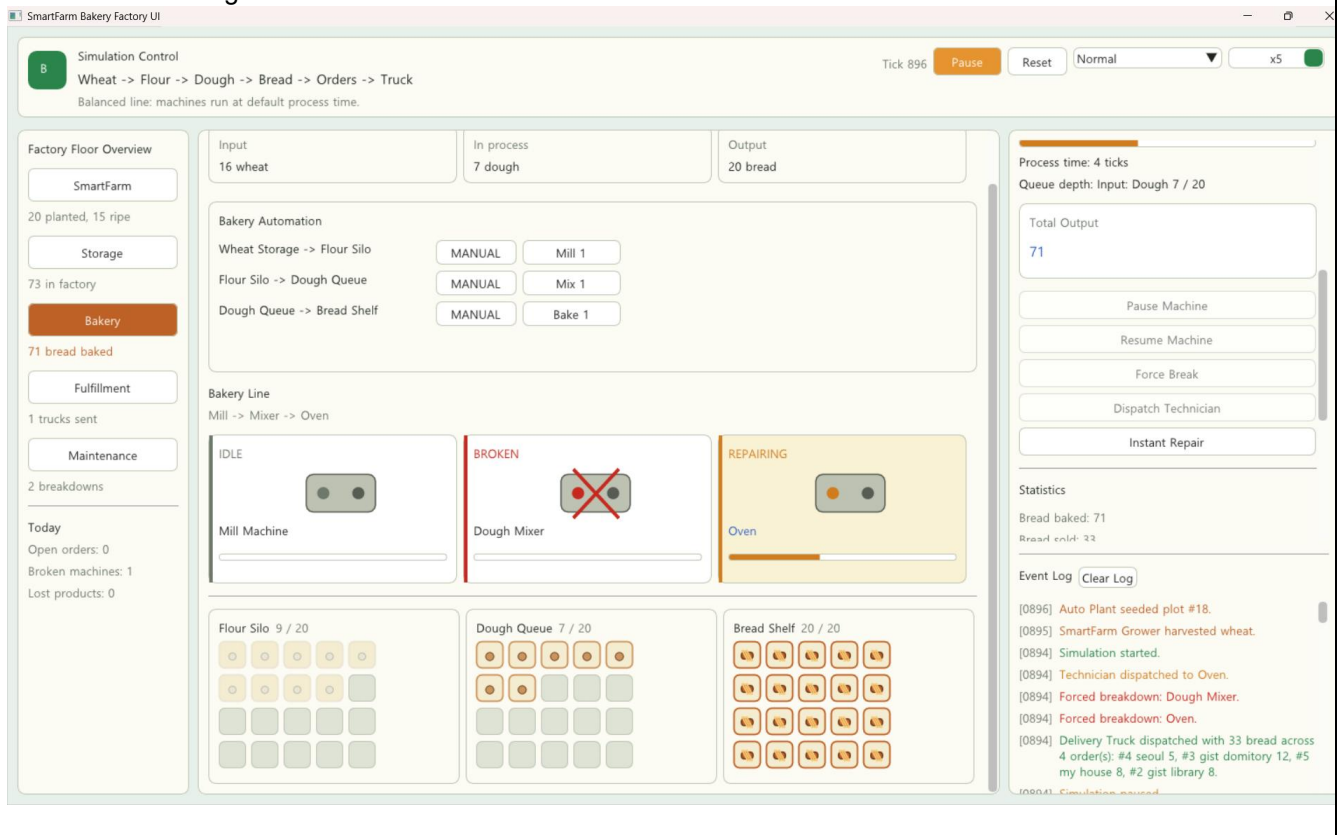
7. Fulfillment view after truck dispatch, showing the increased number of sold bread and delivered trucks.



8. Storage view displaying the current inventory status of Wheat Storage, Flour Silo, Dough Queue, Bread Shelf, and Packed Orders.



9. Maintenance scenario in which machines are forcefully broken and a technician is dispatched to repair the damaged machine.



5. Project Completion Roadmap

#	Task	Owner	Target date	Done?
1	Finalize backend class hierarchy using SimulationObject, Machine, ProducerMachine, ProcessorMachine, ConsumerMachine, and concrete machines.	황서현	May 24	■
2	Implement product hierarchy and storage/order model using Product, StorageBin, OrderItem, and EventItem.	황서현	May 24	■
3	Implement Factory update loop, command handling, and snapshot generation.	황서현	May 24	■
4	Implement Dear ImGui frontend layout with Simulation Control, Factory Floor, Inspector, Event Log, and Statistics panels.	김도경	May 26	■
5	Implement FactoryCommand and FactorySnapshot DTO flow for frontend/backend separation.	김도경	May 28	■
6	Implement bakery production pipeline: wheat, flour, dough, bread, packed orders, and delivery truck.	황서현	May 30	■
7	Implement scenarios: Normal, Bottleneck, Breakdown Manual, Breakdown Auto, and Overflow.	황서현	June 1	■
8	Integrate UI controls with backend simulation actions and machine inspector controls.	김도경	June 3	■
9	Test final demo flow, including order fulfillment, truck dispatch, breakdown, and technician repair.	김도경	June 5	■
10	Finalize UML diagrams, screenshots, report, GitHub repository, and presentation materials.	Joint	June 6	■

6. Instructor Feedback

Progress rating	☐ On track ☐ Minor delays ☐ At risk ☐ Needs discussion
Architecture check	☐ Frontend / backend correctly separated ☐ Needs revision
UML quality	☐ Satisfactory ☐ Incomplete ☐ Incorrect — resubmit
Comments	

Date reviewed	